

(Quy 25) New Formulation for CSApHMI

Features:

- Fixed costs for opening hubs, and for opening hub links
- Incomplete hub network (connectivity)
 - topology constraints on set of Arcs that can be opened
- Multi-level hub capacities. Capacity is a decision variable
- new p-center objective : minimax fraction of added travelling time
- Third objective: minimize the maximum load on any Hub

Bibliography

- On Incomplete networks:
 - (*) The design of single allocation incomplete hub networks (SA Alumur, 2009)
- Surveys:
 - (*) Hub Location Problems, Contreras, 2019 in chapter 12 of "Location Science"
 - Perspectives on modeling hub location problems, (Alumur et al, 2021)
- Capacitated models (with multiple capacity levels):
 - (*) Single-assignment hub location problems with multiple capacity levels
- Foundational Works
 - A Biobjective Hub Location Model with Consideration of Congestion for Railway Transportation Planning (Sunarin 2024)
 - Integer programming formulations of discrete hub location problems (Campbell 1994)
 - (*) Efficient algorithms for the uncapacitated single allocation p-hub median problem (Ernst, 1996)
 - O'Kelly:
 - A quadratic integer program for the location of interacting hub facilities (1987);
 - The Location of Interacting Hub Facilities (1986)

Formulation

MIP - Mixed Integer Programming

Parameters & Assumptions

- $\alpha, \alpha^t, \chi, \delta$: discount factor for transportation cost on hub links, discount factor for travelling time on hub link, collection coefficient, distribution coefficient
- N set of nodes, indexed by $1, 2, 3, \dots, n$

- $H \subseteq N$ set of potential hub nodes, indexed $1, 2, \dots, h$
- $E \subseteq N^2$ set of allowed connection between nodes
 - **Assumption 1:** No direct link between O-D pairs, i.e. $(i, j) \notin E \quad \forall i, j \in N : i, j \notin H$
- $A \subseteq H^2 \cap E$ set of potential hub links
- For convenience, define adjacency matrices e and a as follows:
 - $e_{ij} = 1$ if $(i, j) \in E$, 0 otherwise
 - $a_{ij} = 1$ if $(i, j) \in A$, 0 otherwise
- FL_{ij} fixed cost to open a link between hub $i \in H$ and $j \in H$
- **Assumption 3:** At each potential hub node, there are different levels of hub capacities with different fixed costs
- $Q_k = \{1, 2, \dots, q_k\}$ set of different capacity levels available for a potential hub to be installed at node $k \in H$
- FH_k^q fixed cost for installing a hub with capacity level q at hub k ($k \in H, q \in Q_k$)
- Γ_k^q capacity of a hub installed at node k with capacity level q ($k \in H, q \in Q_k$)
- Preprocess: sort hub levels increasingly in order of capacity, then fixed costs are also increasing.
 - $\Gamma_k^{q_1} < \Gamma_k^{q_2} \quad \forall q_1, q_2 \in Q_k : q_1 < q_2$
 - $FH_k^{q_1} < FH_k^{q_2} \quad \forall q_1, q_2 \in Q_k : q_1 < q_2$
- w demand (flow) matrix, w_{ij} is amount of flow to be sent from node $i \in N$ to node $j \in N$
 - $O_i = \sum_{j \in N} w_{ij}$: flow originating from node i
 - $D_i = \sum_{j \in N} w_{ji}$: flow destined for node i
- t_{ij} time to directly travel between $i \in N, j \in N$.
- d_{ij} distance between $i \in N, j \in N$. Distance matrix d can be computed from t in $O(n^3)$ using Floyd's algorithm
 - **Assumption 2:** d satisfies triangle inequality, i.e.

$$d_{ij} \leq d_{ik} + d_{kj} \quad \forall i, j \in N$$
- c_{ij} transportation cost between $i \in N, j \in N$. In this paper, we set $c_{ij} = t_{ij}$
- d_{ij}, c_{ij}, t_{ij} *closely relates to each other* !
- Question: Whether t_{ij} is symmetric, and thus, c_{ij} too?

Decision Variables

- $x_{ij} = 1$ if node $i \in N$ is allocated to hub at node $j \in H$, 0 otherwise
 - $x_{kk} = 1$ if node k is a hub.
- $z_{ij} = 1$ if a hub link is established between hubs $i \in H$ and $j \in H$, 0 otherwise;
- f_{ij}^k the total amount of flow originating from node $k \in H$ and to be routed on hub link (i, j) in that order $i \rightarrow j$ ($i, j \in H$)
- $h_k^q = 1$ if the hub at node $k \in H$ have the capacity level $q \in Q_k$, 0 otherwise

For p-center objective: build a spanning tree to compute the travel time between hubs in the hub network

- $y_{ij}^k = 1$ if the spanning tree rooted at hub $k \in H$ uses the hub link (i, j) (in direction from hub $i \in H$ to hub $j \in H$; **0 otherwise**
- s_{ij} = travel time from hub $i \in H$ to hub $j \in H$ in the hub network (not discounted), computed from the network topology induced by the decision variables,

Objectives

For brevity, let $\Gamma_k = \sum_{q \in Q_k} \Gamma_k^q h_{kq}$ be the capacity of the hub installed at hub k . Then $FH_k = \sum_{q \in Q_k} FH_k^q h_{kq}$ is the fixed cost for opening this hub.

We have the following objectives:

- Z_1 minimal sum of costs, including fixed cost for opening infrastructure and transportation cost

$$\min \sum_{i \in H} \sum_{j \in H} FL_{ij} z_{ij} + \sum_{k \in H} FH_k x_{kk} + \sum_{i \in N} \sum_{k \in H} (\chi c_{ik} O_i + \delta c_{ki} D_i) x_{ik} + \sum_{i \in H} \sum_{j \in H} \sum_{k \in N} \alpha c_{ij} f_{ij}^k$$

- Z_2 minimax of the fraction of added traveling time between any O-D pairs, in other words

$$\min \max_{i, j \in N} \frac{tH_{ij}}{d_{ij}}$$

in which tH_{ij} denotes the travel time from node $i \in N$ to node $j \in N$ via the H&S network, determined by allocation patterns

$$tH_{ij} = \min_{k, m \in H} t_{ik} + \alpha^t s_{km} + t_{mj} + (2 - x_{ik} - x_{jm})M$$

in the above formula, M is a sufficiently large constant (Big-M method)

- Z_3 is the minmax of passenger load at any hub.

$$Z_3 = \min \max_{k \in H} \frac{\text{total flow incoming at hub node } k}{\Gamma_k}$$

$$Z_3 = \min \max_{k \in H} \frac{\sum_{i \in N} (O_i x_{ik} + \sum_{m \in H: m \neq k} f_{mk}^i)}{\Gamma_k}$$

Constraints

The formulation is subjected to the following constraints for allocation and flow routing

$$\sum_{j \in H} x_{ij} = 1 \quad \forall i \in N \quad (1)$$

$$x_{ij} \leq x_{jj} \quad \forall i \in N, j \in H \quad (2)$$

$$x_{ij} \leq e_{ij} \quad \forall i \in N, j \in H \quad (3)$$

$$\sum_{q \in Q_k} h_k^q \leq x_{kk} \quad \forall k \in H \quad (4)$$

$$z_{ij} \leq x_{ii} + x_{jj} - 1 \quad \forall i \in N, j \in H \quad (5)$$

$$z_{ij} \leq a_{ij} \quad \forall i \in N, j \in H \quad (6)$$

$$\sum_{j \in H: j \neq i} f_{ji}^k + O_k x_{ki} = \sum_{j \in H: j \neq i} f_{ij}^k + \sum_{l \in N} w_{kl} x_{li} \quad \forall i \in H, k \in N \quad (7)$$

$$f_{ij}^k \leq O_k z_{ij} \quad \forall i, j \in H, k \in N \quad (8)$$

$$\sum_{i \in N} \left(O_i x_{ik} + \sum_{m \in H: m \neq k} f_{mk}^i \right) \leq \Gamma_k \quad \forall k \in H \quad (9)$$

for traveling time,

$$\sum_{i \in H, i \neq j} y_{ijk} \geq x_{jj} + x_{kk} - 1 \quad \forall j, k \in H : j \neq k \quad (10)$$

$$\sum_{i \in H, i \neq j} y_{ijk} \leq x_{kk} \quad \forall j, k \in H : j \neq k \quad (11)$$

$$y_{ijk} \leq z_{ij} \quad \forall i, j, k \in H \quad (12)$$

$$y_{ijk} + y_{jik} \leq 1 \quad \forall i, j, k \in H \quad (13)$$

$$s_{kj} \geq (s_{ki} + t_{ij})y_{ijk} \quad \forall i, j, k \in H : i \neq j \wedge j \neq k \quad (14)$$

$$s_{kk} = 0 \quad \forall k \in H \quad (15)$$

let $X_i = \sum_{j \in H} j x_{ij} (i \in N)$ be the index of the hub node which i is allocated to, then

$$tH_{ij} = t_{iX_i} + \alpha^t s_{X_i X_j} + t_{X_j j} \quad \forall i, j \in N \quad (16)$$

Note that we are not imposing any symmetry property on d, t matrix, only the Triangle Inequality.

A main difference compared to Alumur's work is that we do not force hub links to be bi-directional. Therefore, z matrix may not necessarily be symmetric.

The domain constraints are:

$$f_{ij}^k \geq 0 \quad \forall i, j \in H : i \neq j, k \in N \quad (17)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in N, j \in H \quad (18)$$

$$z_{ij} \in \{0, 1\} \quad \forall i, j \in H : i \neq j \quad (19)$$

$$y_{ijk} \in \{0, 1\} \quad \forall i, j, k \in H : i \neq j \wedge j \neq k \quad (20)$$

$$s_{ij} \geq 0 \quad \forall i, j \in H \quad (21)$$

$$h_k^q \in \{0, 1\} \quad \forall k \in H, q \in Q_k \quad (22)$$

Comments

The author (Alumur) noted that the spanning trees constructed not necessarily minimum spanning trees, thus, the distances may not be the shortest distance. However, when optimizing for Z_2 , the variables will tend to regress to the *Shortest Path Tree* and thus achieving true distances.

Solution Methods

Please refer to Slide "...."

Exact methods:

- Branch&Bound
- Invalid inequalities

Meta-Heuristics:

- based on NSGA-II
 - may incorporate the SA algorithm to improve the algorithm, instead of using Local Search
 - references
 - "EFFICIENT ALGORITHMS FOR THE UNCAPACITATED SINGLE ALLOCATION p-HUB MEDIAN PROBLEM" (Ernst'96)
 - Bandyopadhyay, S., Saha, S., Maulik, U., & Deb, K. (2008). A simulated annealing-based multiobjective optimization algorithm: AMOSA. *IEEE transactions on evolutionary computation*, 12(3), 269-283.
 - prototype: the NSMA algorithm, as reported in Week 6
- using the encoding scheme as in "The incomplete hub-covering location problem considering imprecise location of demands" (Davari 2012)
- plan to try out the Solution initialization process as in the papers you gave me
 - 2023 Logistics hub location for HSR freight transport with road-rail intermodal transport network (based on remaining Capacity and Proximity from each hub to a node)
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- Calculation of objective values is trivial. However, time complexity went from $O(n^2)$ to $O(n^3)$
 - We don't know whether imposing more constraints make the feasibility region harder for the search algorithm
 - Need to consider about measures for handling constraints violations